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Advanced Real-Time Capacity Generation Management for Optimized Drilling Operations

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Abstract

Efficient and reliable power management is essential for modern drilling operations, where generator demands are highly dynamic and engine conditions frequently change; conventional generator management, reliant on static heuristics and rated capacities, often leads to suboptimal utilization, excess fuel consumption, and increased maintenance costs and to address these challenges, we present an integrated, real-time system that adaptively estimates generator capacity and optimizes power management in drilling environments¹.

At the core of the system is an Iteratively Weighted Least Squares Regression (IWLSR) model that continuously updates generator capacity using real-time operational data: it's a method that, coupled with dynamic data binning, robust outlier filtering, and continuous model adaptation, delivers significantly improved estimation accuracy, ensuring operational decisions reflect true generator capabilities.

The system also incorporates an artificial intelligence (AI)-driven predictive module employing a Bi-directional Recurrent Neural Network (RNN), trained on extensive rig data to forecast short-term peak power demands several minutes in advance. The RNN's ability to capture nonlinear and non-stationary trends enables proactive generator dispatch, mitigating the risks associated with lagging, reactive responses.

An adaptive generator management framework integrates these estimation and prediction engines, employing a cost-based optimization algorithm that considers fuel efficiency, run hours, cycle counts, and thermal constraints. This approach dynamically selects the optimal generator set, minimizing unnecessary cycling and emissions while maximizing reliability and asset life.

Field deployments across multiple rigs have demonstrated substantial benefits: reductions in generator-related blackouts, accurate early detection of engine anomalies, and fuel savings of 3% to 7%. The adaptive, real-time system is resilient to varying operational conditions, representing a step-change improvement over the state of the art.

This paper details the methodologies, field validation results, and operational advantages of this integrated approach, highlighting its transformative potential for sustainability, reliability, and cost-effectiveness in drilling power management.

Introduction

Efficient and reliable power management in drilling operations is important as it directly impacts costs, emissions, and equipment life; rigs tend to be subject to highly dynamic power demands across activities such as drilling, tripping, and circulation, and conventional generator management systems typically use static heuristics, manual setpoints, or rule-based logic that are, for the most part, are based on rated capacities or conservative derate tables. These types of approaches are unable to capture real-time changes in generator health or rapidly evolving drilling dynamics, frequently resulting in suboptimal utilization, excess fuel consumption, increased emissions, and greater maintenance requirements.

Under controlled conditions, engine load and generator power output exhibit a strong linear relationship, as shown in Table 1 and Figure 1. However, field data from multiple rigs reveal a much more complex reality: the relationship is often only *piecewise linear* and is subject to significant noise from several sources:

- **Regime-dependent noise:** Operational states such as air filter restriction and "cool mode" induce distinct slopes or segments in the load–power trend.
- **Transient dynamic noise:** Rapid load changes during drilling events introduce clusters of outliers.
- **Temporal desynchronization noise:** Load and power measurements are acquired asynchronously from separate systems, causing a smeared fan of points, especially during accelerations.
- **Saturation noise:** Data accumulates along the 100% load boundary due to system limitations.
- **General measurement noise:** Sensor inaccuracies further scatter the data.
- **Quantization noise:** Since the engine load signal is reported without fractional, small variations—especially at low load—are lost. This introduces quantization errors, which act like a floor of noise. In essence, these discretized load readings limit the estimator's ability to distinguish minute changes in capacity, particularly around regime boundaries or during low-load operation.

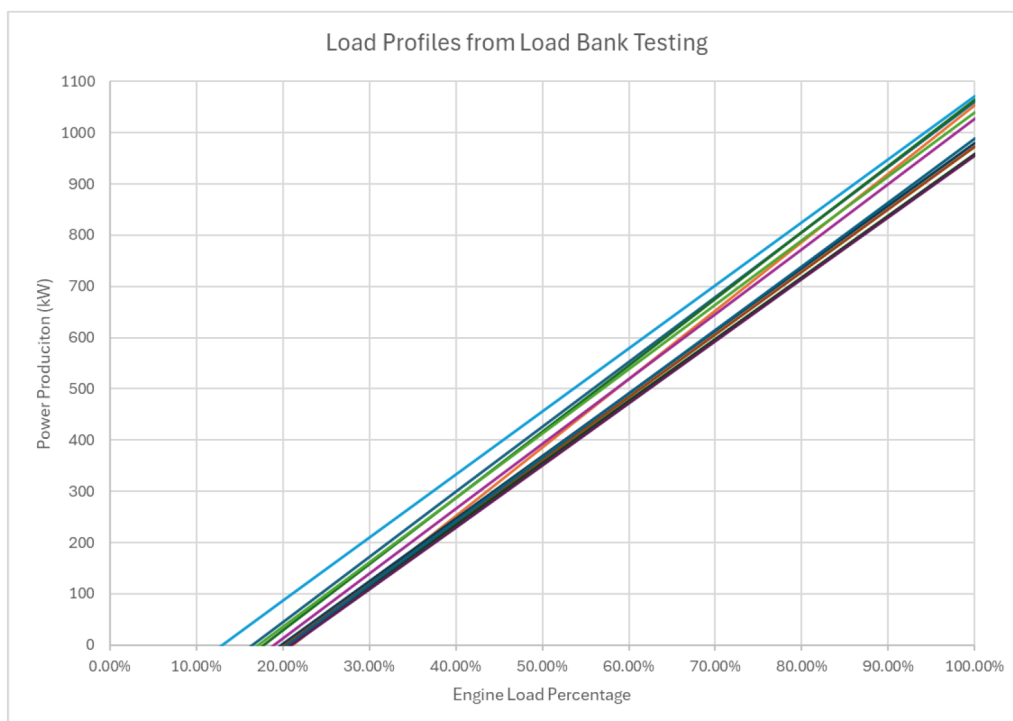


Figure 1—Engine Load vs. Power Output from Load Bank Testing for multiple generators

Table 1—Engine Load vs . Power Output data from Load Bank Testing

LOAD in Kw	14-1156	14-0794	14-1157	14-94079	14-1162	14-94097	14-1276	14-1275	Eng 1	Eng2	Eng3
0	18%	21%	19%	14%	19%	14%	18%	19%			
50	22%	27%	23%	19%	23%	24%	24%	24%	25%	24%	25%
100	27%	31%	28%	24%	28%	28%	28%	29%	29%	29%	30%
150	31%	33%	32%	28%	32%	33%	32%	33%	32%	32%	34%
200	29%	35%	33%	27%	36%	32%	36%	37%	36%	37%	38%
250	33%	37%	34%	30%	41%	35%	40%	41%	41%	40%	42%
300	37%	41%	37%	34%	39%	38%	45%	45%	46%	44%	46%
350	41%	45%	42%	38%	42%	42%	49%	49%	50%	49%	50%
400	46%	49%	46%	43%	47%	47%	53%	53%	53%	53%	54%
450	49%	53%	50%	47%	51%	50%	57%	57%	58%	58%	58%
500	59%	61%	59%	59%	61%	60%	61%	61%	63%	61%	61%
550	63%	66%	63%	62%	66%	64%	65%	65%	67%	65%	65%
600	67%	71%	67%	67%	71%	67%	70%	70%	70%	69%	71%
650	71%	73%	71%	71%	74%	73%	74%	74%	74%	73%	74%
700	76%	75%	74%	68%	78%	76%	77%	78%	80%	77%	78%
750	79%	77%	79%	73%	82%	79%	81%	83%	83%	81%	82%
800	77%	80%	80%	76%	80%	80%	85%	86%	87%	84%	87%
850	81%	84%	81%	80%	84%	83%	89%	90%	90%	88%	92%
900	85%	88%	84%	83%	87%	86%	93%	94%	94%	93%	97%
950	89%	92%	89%	88%	92%	91%	98%	98%	99%	97%	100%
1000	92%	91%	93%	93%	96%	95%	100%	100%			
1050	100%	100%	100%	100%	100%	100%					
Slope	1269.997	1336.435	1296.301	1232.335	1267.718	1256.584	1221.652	1222.068	1206.903	1243.259	1210.362
Intercept	-209.289	-282.936	-231.569	-160.626	-240.817	-215.813	-241.717	-248.371	-247.645	-255.117	-254.246
R ² Value	0.987027	0.990064	0.98961	0.985868	0.98683	0.989231	0.999216	0.999239	0.998977	0.999406	0.998867

These effects are illustrated in Figure 2, which highlights the inadequacy of single-regime or static models for real-time generator capacity estimation. As a result, static generator management schemes cannot provide the reliability or efficiency demanded by modern operations

In practice, these limitations have tangible operational and commercial consequences. Overestimating generator capacity can result in blackouts, engine trips, and equipment failures; underestimating leads to unnecessary generator cycling, excessive fuel consumption, and heightened emissions. These events are not rare; documented field audits have shown unplanned downtime, lost drilling productivity, and regulatory challenges tied to static management. Real-time derating triggers—such as air filter clogs, temperature-induced derates, or gradual capacity loss—are routinely missed by traditional systems, resulting in avoidable maintenance and cost overruns.

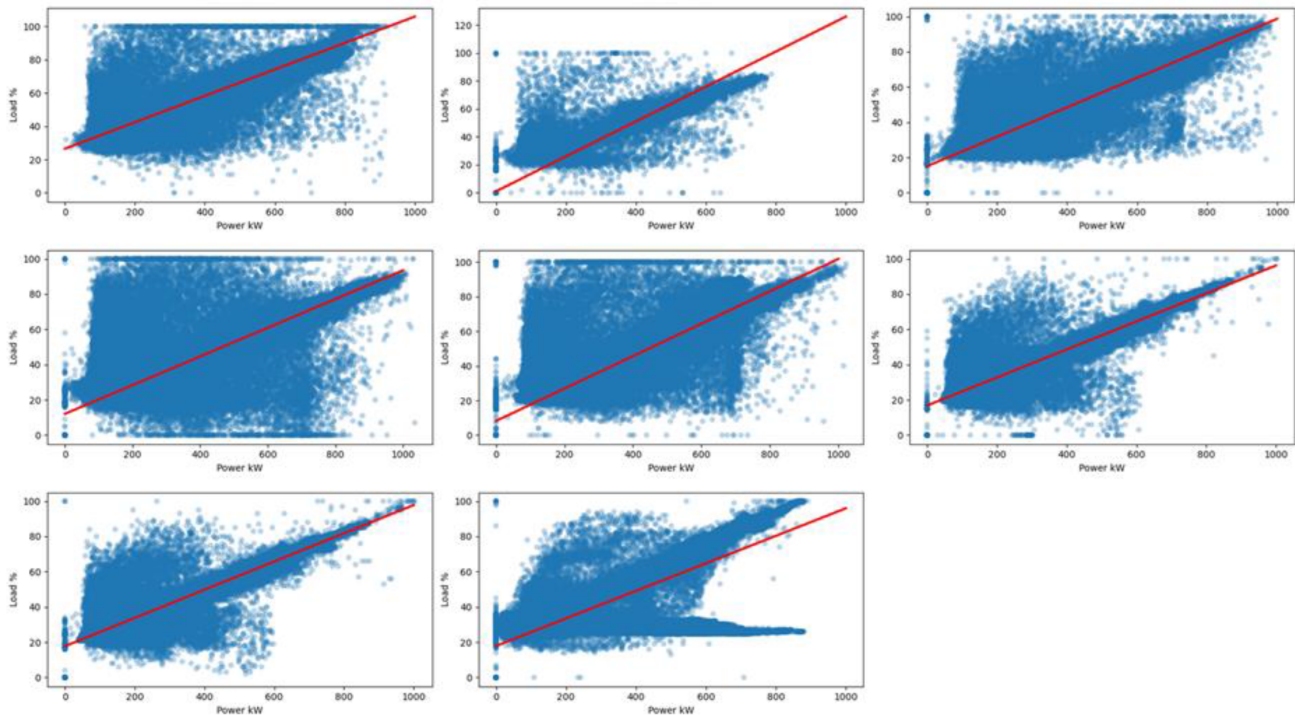


Figure 2—Field data showing diverse noise phenomena in the engine load vs. power relationship across multiple engines and rigs. Distinct piecewise linear regimes, transient smearing due to temporal misalignment, saturation at 100% load, and general sensor noise are evident.

While prior work has demonstrated the promise of AI-driven load prediction and condition monitoring (Koritala et al. 2023), no fielded solution has provided robust, adaptive generator capacity estimation capable of handling the noise and regime shifts typical in drilling operations. This paper addresses this critical gap by presenting an advanced system that integrates real-time generator capacity estimation, predictive power management, and cost-based optimization for generator selection.

At the core of this solution lies an adaptive Iteratively Weighted Least Squares Regression (IWLSR) algorithm, which enhances capacity estimation by dynamically assigning weights to recent data, thereby improving resilience against outliers and operational irregularities. Complementing this, a Bi-directional Recurrent Neural Network (RNN), as detailed by Koritala et al. (2023), forecasts short-term peak power demands with high temporal sensitivity, facilitating anticipatory generator dispatch and load balancing. These two modules—estimation and prediction—are integrated within a real-time optimization framework that continuously assesses all feasible generator configurations. The system strategically selects combinations that minimize operational expenditures, including fuel usage, runtime duration, start-stop cycles, and thermal degradation penalties, while maintaining reliability and extending equipment lifecycle.

This integrated, adaptive framework demonstrates significant advancements over traditional methodologies, optimizing generator use, fuel efficiency, and emissions while minimizing maintenance and cycling. The following sections detail the system methodology, field validation, and operational benefits achieved in multiple deployments.

Methodology

Generator Capacity Estimation via Adaptive IWLSR

Field conditions on drilling rigs demand a generator capacity estimation system that is not only robust to diverse and unpredictable noise sources, but also agile enough to adapt in real time to abrupt regime changes and the gradual effects of wear and degradation. Evidence shows that the harsh operational environment

—marked by sensor drift, asynchronous data, mechanical failures, and human interventions—means any static, batch, or naive online estimator will inevitably fail or lag behind reality as illustrated by Figures 3 and 4. To solve this, a layered methodology is used that combines overlapping steady-state detection, aggressive real-time data pruning, and a rotating bin buffer, feeding an Iteratively Weighted Least Squares Regression (IWLSR) model with dynamically adaptive weighting. Pruning is also implemented by ranking and removing outlier points across all bins where the data no longer reflects the current process. Finally, an active memory bank is used for model selection and merging, allowing instant regime switching and graceful degradation or recovery, ensuring estimates remain both physically plausible and operationally actionable, no matter how field conditions evolve.

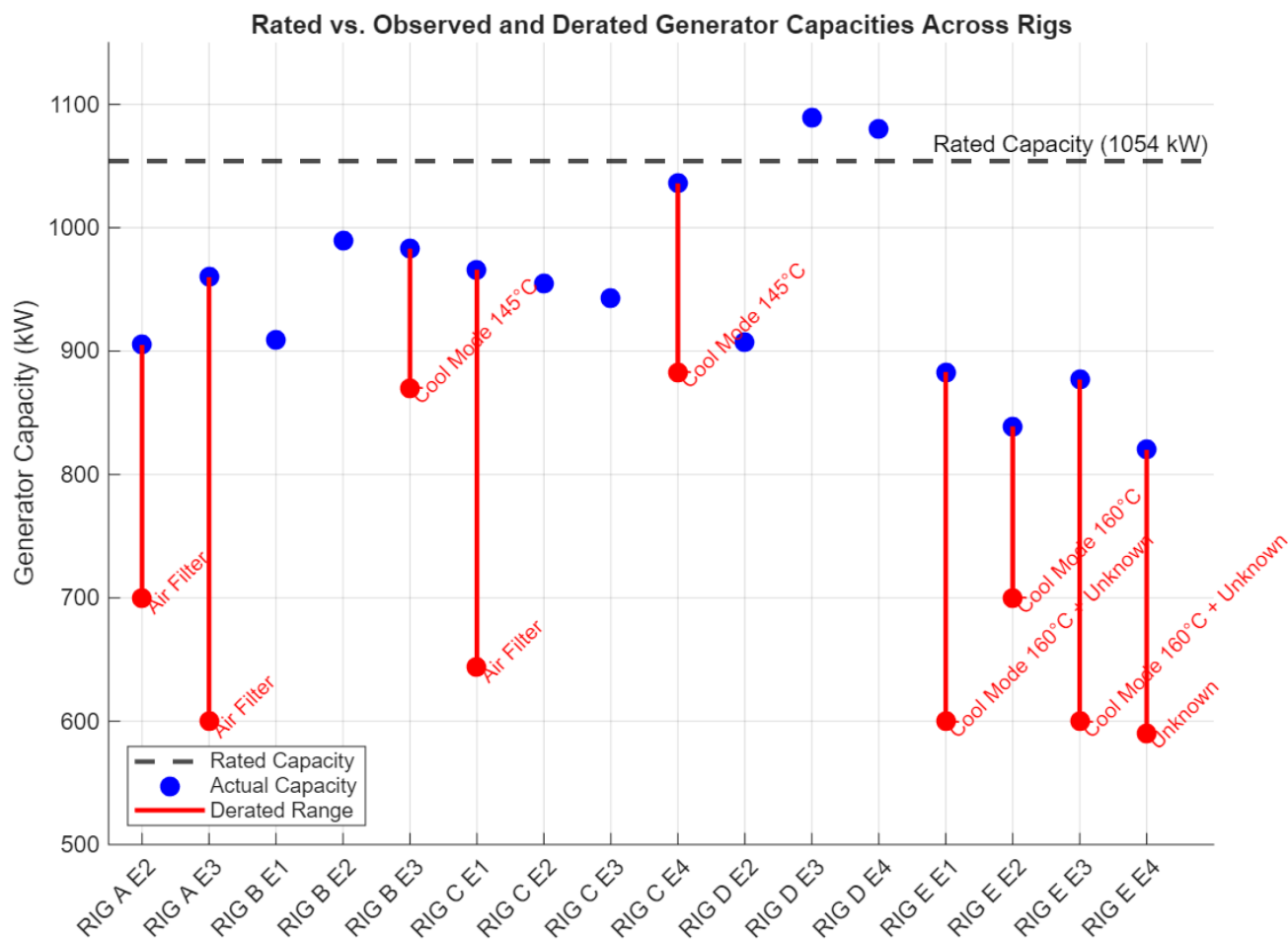


Figure 3—Rated versus actual generator capacities across multiple drilling rigs, highlighting significant variability and derating events. Despite identical engine ratings (1054 kW), real operational capacities vary widely.

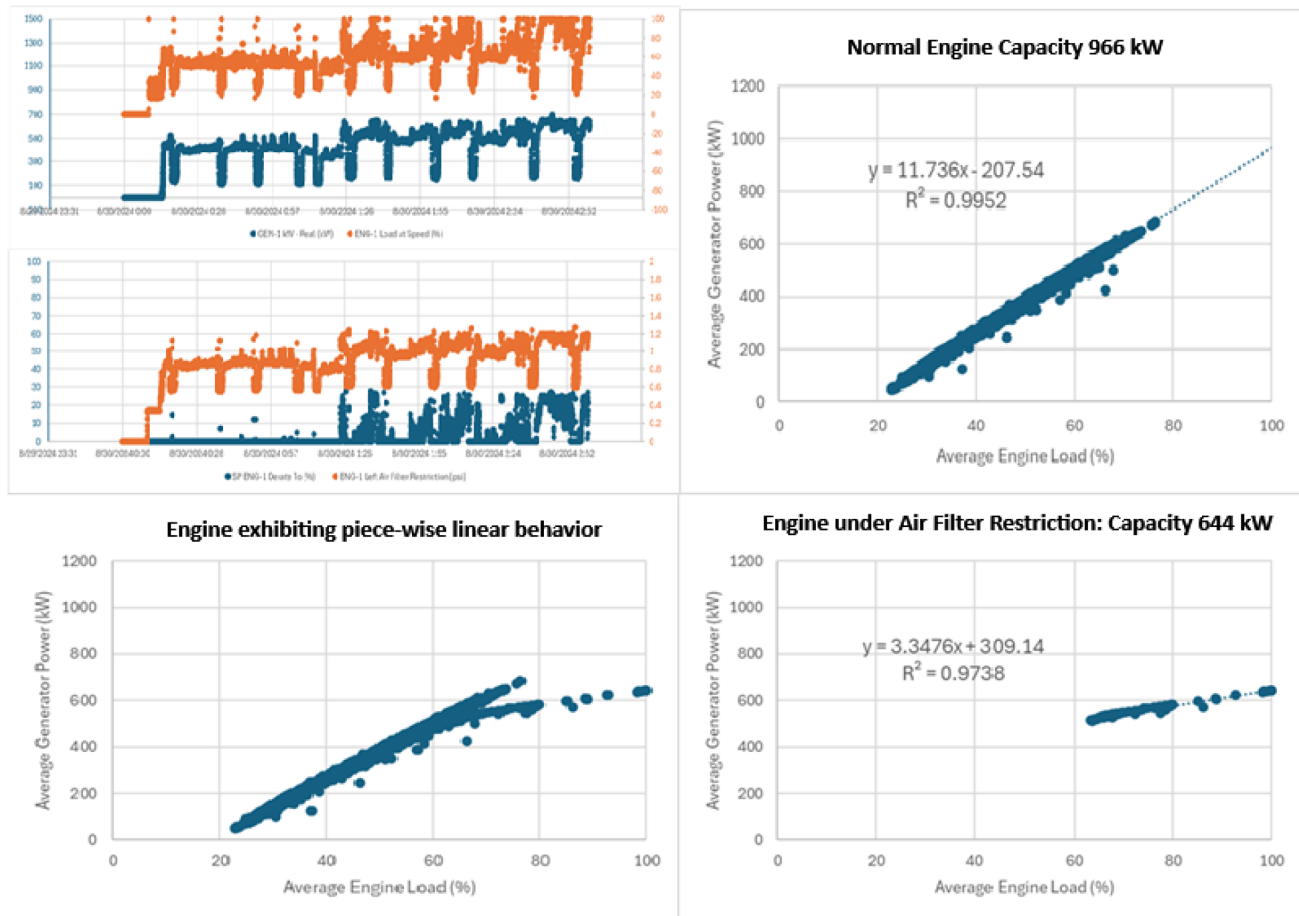


Figure 4—Progressive Engine Derating Due to Air Filter Restriction for a Single Engine. Generator power and engine load (top-left), alongside the increase in air filter restriction and the resulting derate. Regression plots (right and bottom-left) illustrate the shift from normal operation (966 kW) to severe derating (down to 644 kW), revealing the development of piece-wise linear relationships between load and power as air restriction increases.

Data Preprocessing and Noise Mitigation. The first challenge in a real-world implementation of any machine learning algorithm is to reliably isolate "good" data—steady-state, representative of true generator capacity—from process noise, asynchronous sampling, or operational upsets. One drilling connection or block move can contaminate seconds or minutes of data, an engine can also suddenly derate past certain operational points, and the data acquired from different sources will be misaligned.

To mitigate these issues, we apply **overlapping time-window averaging** for both generator power and engine load data. Each data point contributes to multiple overlapping windows, ensuring that transient disturbances are effectively averaged out, thus eliminating the need to align them as the window is made much larger than the delay. This method also avoids the risk of discarding valuable steady-state information caused by isolated disturbances. For instance, a momentary spike in load due to a brief operational upset would not be sufficient to discard an entire window of data, as only the transient parts of the signal are affected, leaving the steady-state data intact.

Following the computation of these data windows, a **rigorous statistical validation protocol** is applied to ensure that only high-integrity data contributes to the capacity estimation process. The initial screening step assesses whether the power and load measurements adhere to predefined variability thresholds. Specifically, the **standard deviation** of these signals is computed and compared against acceptable bounds to **detect instability**. Windows exhibiting excessive fluctuations are flagged as invalid, as they likely stem from transient system behaviors, such as sensor malfunctions or external disturbances; for

instance, pronounced power oscillations within a window typically indicate transitional states or operational anomalies, rendering such data unsuitable for reliable capacity inference.

In addition to ensuring acceptable variability, we **reject data points where there is an excessive rate of change in power or load** between consecutive samples. Abrupt changes in power or load—such as a sudden jump in load caused by operator intervention—are indicative of transient events rather than steady-state operation. To account for this, we compute Δpower and Δload for every overlapped window, using those gradients to screen out rapidly varying samples. If the change exceeds a predefined threshold, the data is flagged and excluded from the regression model. **This ensures that only data from stable operational phases is used**, preventing the model from being influenced by short-lived transients.

Finally, we **apply minimum thresholds for both average power and load**: low-power windows (where the generator is idling, warming up, or offline) and loads **outside the normal operating range** are excluded. These measures collectively ensure that only stable, representative data—indicative of true operational capacity—are logged into bins for model estimation. Figure 5 illustrates this process, highlighting how robust preprocessing separates valid capacity data from noisy or unstable conditions.

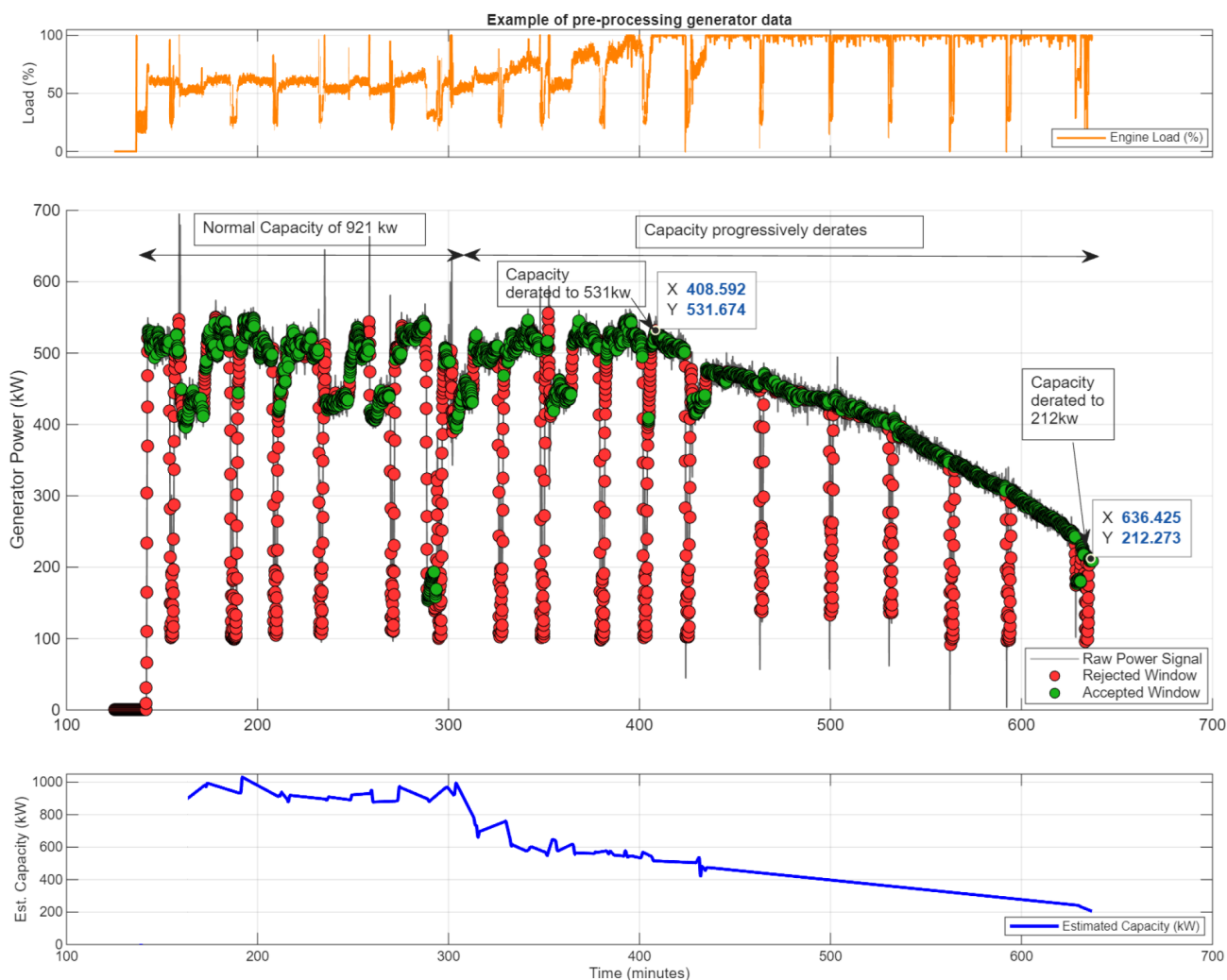


Figure 5—Generator data preprocessing and adaptive capacity estimation. Top: Engine load (%) over time. Middle: Raw generator power (black) with color-coded window means—green (accepted), red (rejected) based on steady-state criteria. Bottom: Real-time estimated generator capacity (blue), clearly illustrating progressive derate onset around minute 320. Robust data filtering and windowing enable the estimator to track both abrupt and gradual changes in generator performance.

Dynamic Binning and Post-Fit Pruning for Model Accuracy. Once valid steady-state data events are isolated through rigorous preprocessing, the subsequent phase focuses on structuring these events for real-time regression modeling: to support both adaptive responsiveness and modeling robustness, the system leverages a **dynamic rotating bin buffer** architecture. This mechanism **facilitates continuous refinement of the regression model** by persistently incorporating fresh, high-fidelity data while systematically retiring stale or contextually irrelevant information. The buffer thereby ensures the model remains attuned to evolving operational conditions without being burdened by outdated behavior patterns.

Each validated data point is sorted into one of **equally spaced bins**, spanning the generator's operational load range. These bins function as memory segments, each capturing a narrow window of the load spectrum (e.g., 60–65%, 65–70%, etc. Fig. 6). With this approach we guarantee **broad coverage across the engine's operating envelope**, preventing the regression from being skewed towards clusters in any region (such as idle or full load).

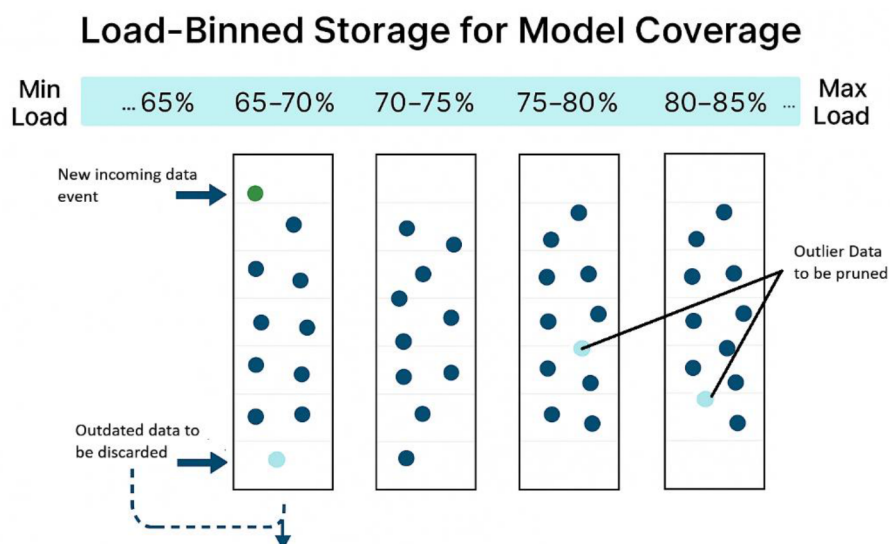


Figure 6—Dynamic Load-Binned Buffer for Real-Time Capacity Modeling. Steady-state data points are sorted into rolling buffers by load bin, ensuring broad operational coverage. New data (green) replaces the oldest (light blue), and outlier points are selectively pruned based on model residuals.

Each bin holds a **fixed-length rolling buffer** where older data points are discarded as new ones arrive. This design provides temporal responsiveness while ensuring stability: we retain enough history to smooth out fluctuations but avoid accumulating stale or irrelevant data. This is particularly important in drilling applications where engine load patterns often shift due to formation transitions, hole cleaning cycles, or top drive activity.

Next, to preserve model quality over time, we implement a **post-regression pruning mechanism** based on the **Median Absolute Deviation (MAD)** of residuals. After enough samples have been accumulated across bins (to prevent early instability), we periodically scan the entire buffer for outliers. Specifically, we compute the MAD of the residual errors and flag any point exceeding a scaled threshold for removal - this pruning serves two purposes:

- **It removes stale points** that may have been valid when inserted but no longer align with the evolving model, such as data from before a regime change (e.g., before air filter replacement or thermal derate).
- **It eliminates poor-quality data** that may have slipped through preprocessing due to sensor anomalies or momentary misbehavior, which are only detectable once a consistent model has emerged.

So, **pruning is performed at the buffer level** and leverages the statistical coherence of the fit to identify and remove structural inconsistencies, refining the bin contents based on how well it aligns with the live model. This mechanism ensures that the model converges quickly during stable periods and can respond effectively to changes in generator condition.

Then, only after a **minimum number of bins** contain valid data, and each has enough samples, do we declare the system ready for regression. This constraint ensures that our model is based on data that spans multiple load conditions, not just narrow operating modes.

Iteratively Weighted Least Squares Regression (IWLSR): Model Fitting with Residual-Aware Refinement. Once the buffer has accumulated a sufficient distribution of high-quality validated data points, the estimator transitions into **active model fitting** using an Iteratively Weighted Least Squares Regression (IWLSR) routine. This stage transforms the current state of the system into a linear model of generator capacity with respect to engine load, explicitly accounting for signal noise, sensor misalignment, and operational variability. To start, we assume a linear relationship between engine load percentage x_i and generator power output $\hat{y}_i$:

$$\hat{y}_i = Ax_i + B$$

Where:

- $\hat{y}_i$ is the predicted power output for sample i ,
- A is the slope (capacity per % load),
- B is the intercept (baseline power at 0% load).

Then the predicted **capacity at 100% load** is given by:

$$P_{cap} = A * 100 + B$$

Unlike basic linear regression, which can be affected by noisy measurements, outliers, or stale data, the **IWLSR approach is more selective in how it treats each data point**. It gives more weight to points that fit the current trend and naturally prioritizes newer information, while downplaying those that don't align (the outliers or older samples). This means that data points that disagree with the evolving model have less impact over time, making the capacity estimate more robust and responsive. The model does this by **assigning a specific weight to each data sample**, w_i , which scales the contribution of each data point to the model using the following relationship:

$$J(A, B) = \sum_{i=1}^N w_i \cdot (y_i - (Ax_i + B))^2$$

This function is then minimized following an iterative procedure:

1. **Initialize:** Perform unweighted least squares regression to get A_0, B_0 .
2. **Residuals:** Calculate $r_i = y_i - (A_k * x_i + B_k)$ at iteration k .
3. **Weight Update:** Recompute w_i using residuals and timestamps.
4. **Weighted Regression:** Solve the weighted normal equations for $A_{\{k+1\}}, B_{\{k+1\}}$.
5. **Convergence Check:** Stop if WMSE does not improve or $k_{\max}$ is reached.

This process repeats, gradually refining the model until it converges on a solution and outliers no longer have a major impact. During the iteration, quality measures like the **coefficient of determination (R^2)** and **mean squared error (MSE)** are used to **help the estimate stay accurate and reliable** for real-time decision-making.

This approach **makes the estimator robust not only to the various sources of noise** outlined above, but to real-world equipment issues. For example, if power output drifts over time—say, because of a filter starting to clog (see Figs. 4 and 5)—the model quickly adapts by shifting more weight to recent, representative points, all without manual intervention.

While the model will adapt naturally over time, **its responsiveness is critical** when subject to sudden derating events, as this will lead to overestimating capacity in the interim, risking a blackout. This is especially true of equipment derating events such as cool mode (Fig. 7), which happen abruptly, but with very consistent historical behaviors that can be modeled. **To address this, we introduce model memory and regime switching.**

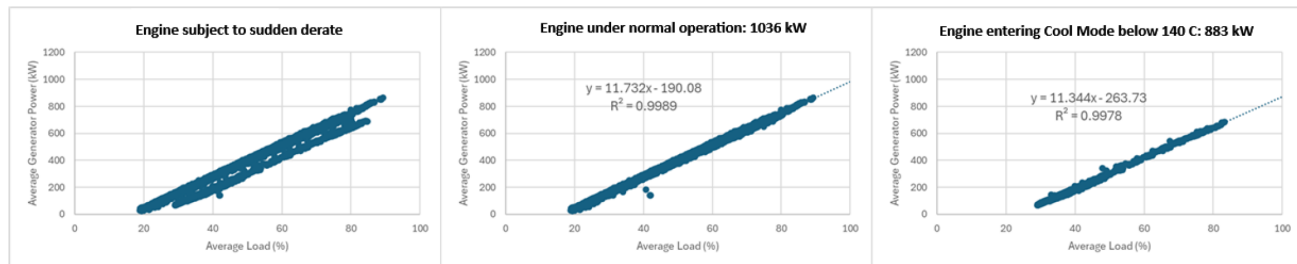


Figure 7—Dual-Mode Linear Capacity Relationships for an Engine showing average generator power versus engine load under varying temperature regimes. Two distinct linear relationships emerge, corresponding to operation in "cool mode" and normal conditions. The reduced capacity during cool mode (883 kW) is due to a lower-temperature thermostat limiting maximum output until engine temperature exceeds 140°F, after which full capacity (1036 kW) is restored.

Model Memory and Regime Switching: Maintaining Operational Continuity Across Capacity States. The output of the IWLSR process is only as useful as its consistency under changing conditions. In drilling operations, generator behavior is not stationary—cooling system events or maintenance resets can all introduce sharp discontinuities in the load-to-power relationship. To address this, our estimator implements a **model memory and switching mechanism** that ensures robust and reactive capacity estimation across distinct operational regimes.

While IWLSR adapts rapidly within a given regime, transitioning between regimes can introduce instability or delay. For instance:

- If a generator enters "cool mode" due to a perceived low engine temperature, the active model quickly becomes invalid—without a fallback, the system must relearn from scratch (Fig. 7).
- After maintenance (e.g., filter replacement), a derated model may no longer reflect reality, but stale points can linger long enough to mislead the estimator (Fig. 12).

To avoid these pitfalls, we store a set of **previously learned, high-confidence models**, each representing a unique capacity state derived from distinct operating conditions. These models are constantly re-evaluated for relevance and quality.

At any point in time, the estimator maintains **historical models**, each defined by:

- Slope A_m , intercept B_m (line fit)
- Associated quality metrics: R_m^2 , MSE, and usage count
- Last validation timestamp

When IWLSR becomes unstable or insufficient data is available to converge, the system computes the **prediction error** of each stored model against the current filtered data point:

$$e_m = |y_{current} - (A_m \cdot x_{current} + B_m)|$$

The model with the **smallest residual error**, provided it passes configured quality thresholds, is selected and used as the active capacity model. This ensures that even during transitions or data droughts, the system can produce a valid, bounded estimate.

However, over time, different operating states may produce statistically indistinct models. To reduce redundancy and noise in long-term memory, the estimator computes the **integral difference** between all stored models:

$$I = \int_{x_{min}}^{x_{max}} (A_i x + B_i) - (A_j x + B_j) dx$$

If $I < \epsilon$, where ϵ is a similarity threshold, the models are **merged** using a weighted average of parameters. Conversely, models that consistently fail to match new data, predict unrealistic capacities, or drift far from any current behavior are **pruned** from memory.

While this approach gives us a clear picture of what each generator can safely deliver, effective power management also requires anticipating future load spikes. This forecasting capability is critical for ensuring that generator dispatch decisions are both timely and reliable, minimizing the risk of blackouts and unnecessary cycling. So, the following section describes our approach to short-term peak power prediction and its integration into the overall management system.

Peak Power Prediction

Real-time capacity estimation is the foundation of safe, efficient generator management, but on its own, it isn't enough to ensure operational security. In drilling, the most severe and **costly failures are often triggered not by gradual trends but by short-lived, dramatic surges in power demands**, such as those occurring during pipe connections, equipment ramp-up, or sudden operational changes. These activities can be highly variable, driven by a combination of equipment differences, operational practices, and unpredictable events, but they are **often very repeatable and therefore predictable**.

So, to truly enable proactive and robust generator dispatch, the management system must not only **track current capacity but also anticipate when and how load will peak**. Both components are essential to predicting imminent power requirements to minimize blackouts, optimize equipment utilization, and reduce unnecessary generator starts.

To that end, an AI-driven estimator capable of predicting power requirements up to 10 minutes ahead was developed as described in [Koritala et al. 2023](#). Our predictor is a **bi-directional LSTM—an RNN architecture** designed for multivariate time-series—chosen for its ability to learn forward- and backward-temporal dependencies. The model **was trained on (1 Hz) data from over 200 wells** across diverse drilling environments, equipment fleets, and operational practices, ensuring broad applicability and robustness.

Key features of the approach include:

- **Multivariate input:** The model continuously ingests a sliding window of process and power data—including bit depth, block position, hookload, pump pressure, and other process KPIs—capturing the full operational context as shown in [Figure 8](#).
- **Proactive prediction:** Every second, the system predicts the peak power for the next 10 minutes, allowing the generator manager to proactively issue start/stop commands and maintain an optimal reserve margin.
- **Blackout-averse training:** The model was influenced to penalize under-prediction (blackout risk) more heavily than over-prediction.
- **Operational resilience:** Data cleaning, interpolation, and specialized heuristics ensure that the model remains stable and reliable even during data interruptions or rare operational modes (e.g., reaming, tripping, or extended pauses).

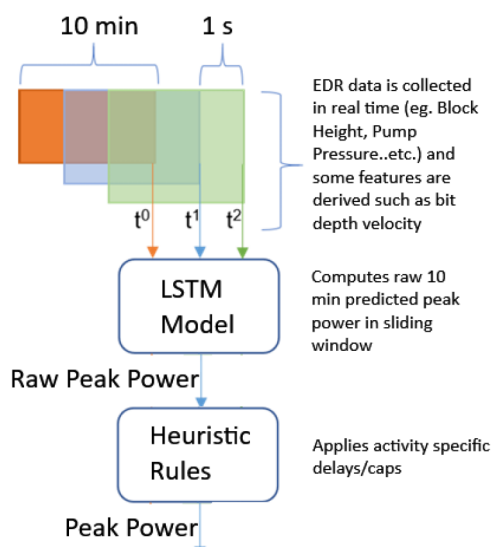


Figure 8—Overview of the peak power forecasting workflow, combining real-time EDR data, an LSTM-based prediction model, and post-processing heuristics to generate actionable peak power estimates.

The approach to training this model made it adaptable to all types of operations on a drilling rig, like drilling, tripping, and casing, even the less frequent ones. However, to maintain a stable output, some post-processing heuristics had to be incorporated. They maintain a smooth estimate and make it more robust to any sudden change or interruption. These include maintaining previous peak predictions during any pauses in drilling activity to be ready when operations continue, as well as handling conditions such as reaming/circulating, which are less frequent drilling activities.

As shown in Figure 9, this approach transforms generator management from a reactive "wait and see" exercise into a proactive, risk-aware control system. By looking ahead to expected demand surges, the system can avoid the "all engines online" inefficiency common in legacy operations, minimizing unnecessary starts and stops while still protecting against blackouts.

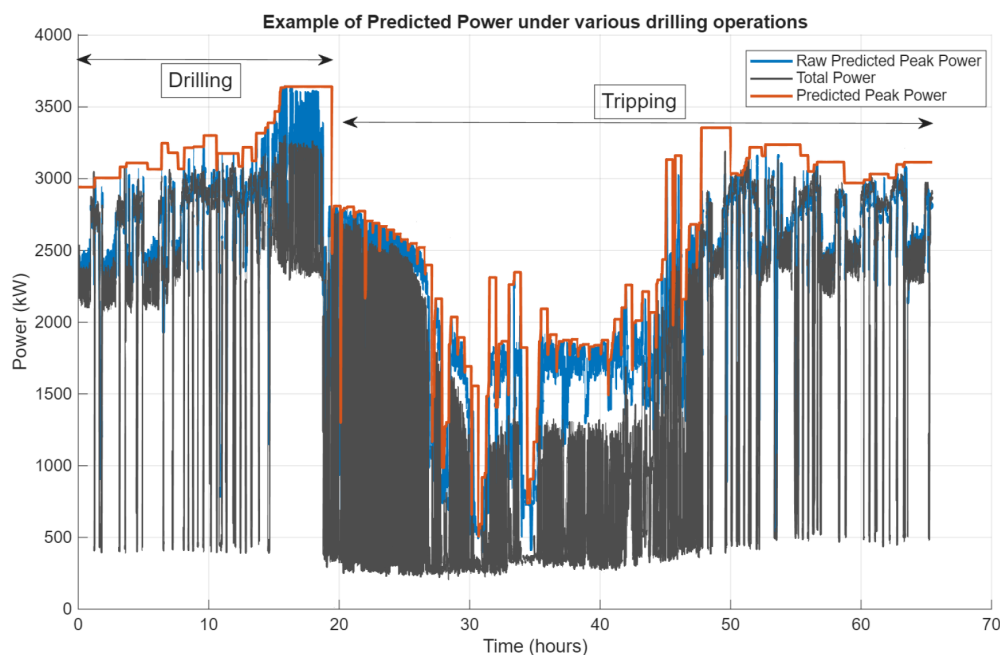


Figure 9—Example of predicted peak power versus measured total power during various drilling operations.

Generator Management and Dispatch Optimization

Traditional generator management approaches often rely on static, heuristic-driven, or manual intervention, but they struggle to handle rapid load fluctuations, sudden derating events, or equipment health changes characteristic of drilling environments. To overcome these limitations, we have developed a dynamic, multi-objective optimization framework that continuously adapts to operational demands, generator conditions, and environmental constraints, significantly enhancing efficiency, reliability, and responsiveness (Fig. 10).

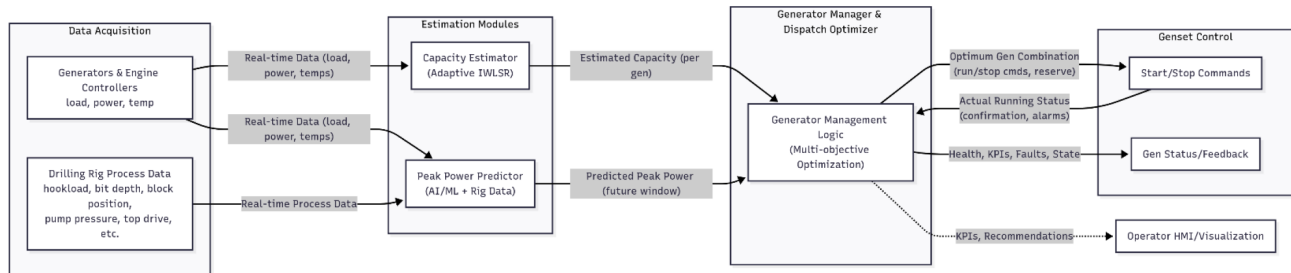


Figure 10—Block diagram of the adaptive generator management system. Real-time data like load and power from generators and engines are combined with rig process data to estimate derated generator capacity and near-term peak power demand. These estimates feed a multi-objective management logic that selects the optimal generator set to run, issuing start/stop commands and monitoring feedback in real time.

Overview and Objectives. The adaptive generator management system has one primary goal: **determining the optimal set of available generators that ensures the predicted peak power demand (plus reserve) is consistently met based on available capacity.** However, for the solution to be optimum, it should be complemented by several other objectives:

- **Performance:** The combination selected should be based on the **lowest number of generators needed to meet the demand with the highest available capacity.** So, rather than running multiple generators at partial loads, this favors higher utilization of fewer units, thereby **maximizing fuel efficiency, lowering emissions, and reducing operational costs.**
- **Equipment Life:** The operational load and start/stop cycles should be **distributed more evenly across all available generators** as it prolongs equipment life, reduces maintenance frequency, and enhances overall reliability.
- **Robust:** As the system is mission-critical, it must be robust, which is achieved by incorporating real-time capacity estimates and predictive load modeling. This allows it to **rapidly adapt to unforeseen events** such as sudden equipment derates, overheating scenarios, or abrupt changes in drilling dynamics.
- **Reliable:** To prevent operational instability, the system incorporates deliberate stabilization mechanisms, including **cooldown periods, anti-cycling timers, and steady-state costs that actively discourage frequent combination changes.** These constraints provide a stable operational environment, maintaining continuity even under fluctuating load demands and transient disturbances.
- **Flexible:** The framework is inherently flexible, **allowing the prioritizing of certain objectives explicitly through adjustable weights.** Whether the priority is minimizing fuel usage, balancing mechanical wear, avoiding generator overheating, or ensuring redundancy, these objectives can be tuned explicitly.

To balance these objectives, a cost-based optimization routine is introduced that dynamically identifies the optimal combination of generators to meet current and projected conditions based on real-time data; these include predicted peak load demands, generator capacity, health and availability as well as certain performance metrics.

Generator Combination Generation and Cost Evaluation. The heart of this adaptive generator management strategy lies in its ability to systematically evaluate and rank all feasible combinations of available generators according to multiple performance criteria. This approach moves beyond static or heuristic-driven selections, incorporating real-time information to ensure robust, optimal decisions. Each decision cycle follows a structured yet dynamic methodology:

At every decision interval, the system first generates all potential combinations of available generators. Combinations are immediately disqualified if they:

- **Fail to provide sufficient reserve power:** Each considered set must meet or exceed predicted peak load, plus a configurable reserve margin based on the estimated capacities of each available generator.
- **Contain unavailable, overheated, or uncontrollable generators:** Any unit flagged for maintenance, experiencing overheating conditions, or previously marked as unavailable is automatically excluded but accounted for.
- **Violate operational constraints:** Combinations that fall below a minimum count of active generators required for redundancy and operational security are also excluded from further consideration.

For the remaining valid combinations, a cost-driven optimization is applied. The optimization process translates multiple operational considerations into quantifiable cost terms, each reflecting critical operational trade-offs. These cost components can include:

1. **Generator Count Cost (Fuel Efficiency):** Combinations with fewer generators are prioritized to minimize fuel usage.

$$GenCost = W_{gen} \times \frac{GenAvailable}{GenOnline}$$

2. **Capacity Utilization Cost (Load Efficiency):** Optimal combinations maximize the loading efficiency of fewer generators, encouraging operations closer to their rated capacities.

$$CapacityCost = W_{cap} \times \left(1 - \frac{CombinationCapacity}{TotalAvailableCapacity}\right)$$

3. **Operational Hours Balancing Cost (Maintenance and Reliability):** Penalize combinations that disproportionately use generators with high accumulated run hours, promoting a balanced distribution of usage and reducing long-term maintenance issues.

$$HourCost = W_{hours} \times \text{sigmoid}\left(CombinationOnHours - \frac{CombinationOffHours}{OperationalBalanceFactor}\right)$$

4. **Cycle Cost (Equipment Wear):** Frequent cycling (start/stop operations) accelerates mechanical wear. Thus, the cost is higher for combinations that would result in frequent cycling of certain units.

$$CycleCost = W_{cycle} \times \frac{CombinationCycles}{TotalCycles}$$

The combined optimization cost is then minimized and succinctly represented as:

$$TotalCost = GenCost + CapacityCost + HourCost + CycleCost + OtherCosts \dots$$

Here, each weight term is independently configurable, allowing operators and system designers to adjust the optimization balance according to operational priorities such as fuel efficiency, equipment longevity, or reliability. By structuring the decision process around clearly defined cost functions, the system achieves consistent and transparent decision-making, significantly enhancing generator management efficiency and robustness.

Command Issuance and Operational Safeguards. After determining the optimal generator set through cost-based optimization, the system transitions into an execution phase, carefully issuing start and stop commands to implement the selected generator combination. Given the critical nature of power generation in drilling operations, this command issuance stage incorporates multiple layers of operational safeguards, ensuring robustness, stability, and reliability, which is illustrated in Figure 11, where brief fluctuations in generator recommendations are ignored unless sustained, preventing excessive cycling during load transitions.

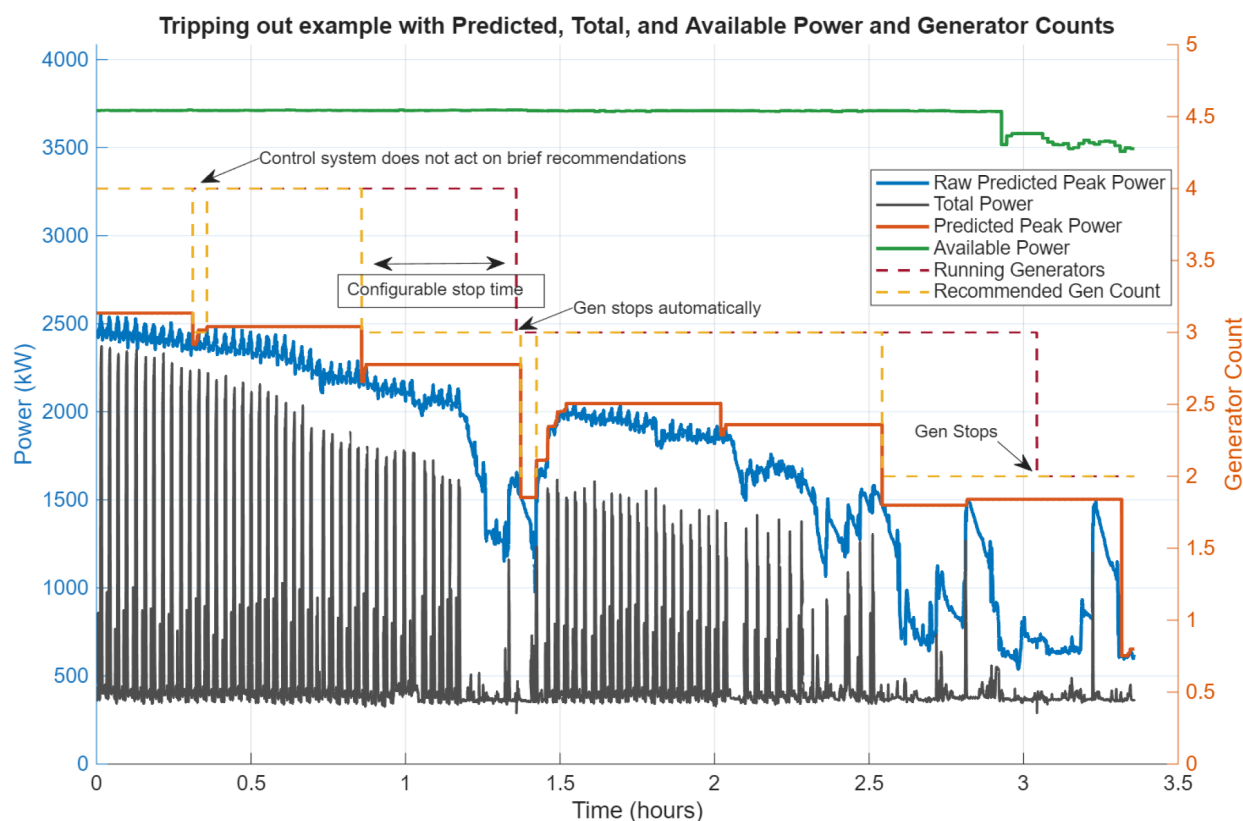


Figure 11—Adaptive generator management during tripping out. Shown are predicted peak power (blue), total rig power (black), available generator capacity (green), the number of running generators (red dashed), and recommended generators by the optimizer (orange dashed). Control logic is shown to ignore brief, transient recommendations in favor of stability thanks to built-in anti-cycling and configurable stop delays. As the predicted and actual load decreases, the system automatically phases off generators in response to real-time power requirements and reserve constraints, ensuring that the generator fleet remains closely matched to operational demand.

To optimize both mechanical health and operational stability, the system adheres to specific rules for starting and stopping generators:

- **Start Commands:** When additional generation capacity is required, the system issues synchronized start commands to all necessary units simultaneously; it's a parallel start strategy that ensures rapid achievement of target capacity, minimizing transient power disturbances and promoting immediate stability under load changes.
- **Stop Commands:** Conversely, generator shutdowns are carefully staged, occurring sequentially rather than simultaneously. By stopping only one unit at a time, the system maintains redundancy, safeguards against unexpected load spikes, and avoids excessive transient conditions.

To protect equipment from excessive mechanical stress and thermal fatigue, robust timing mechanisms and cooldown periods are integrated into the control logic:

- **Anti-Cycling Delay Timers:** Start and stop commands are subject to configurable delays, effectively preventing rapid generator cycling. Such frequent cycling accelerates wear and reduces component life; thus, introducing deliberate delays ensures mechanical components experience stable operational periods.
- **Cooldown Enforcement:** After any generator is identified as overheated or nearing thermal limits, the system places it into a deprioritized status, prohibiting its immediate reuse unless there is an immediate power demand that cannot be met. This enforced cooldown period allows thermal recovery, extending equipment life and maintaining operational reliability.

Reliability in generator management critically depends on robust detection and handling of equipment failures or malfunctions:

- **Failure Detection and Automatic Removal:** Each generator's responsiveness to start and stop commands is monitored continuously. If a generator fails to respond within a specified interval, it is flagged as unavailable, automatically excluded from future combinations until manually or programmatically reset. This approach ensures faulty or failed units do not compromise system reliability.
- **Command Retry Mechanisms:** If initial start or stop commands fail, the system automatically initiates limited retries. This redundancy mechanism significantly enhances operational reliability, ensuring transient communication issues or temporary faults do not result in unnecessary unit downtime.
- **Redundancy Enforcement and Capacity Safeguarding:** The system strictly enforces minimum online generator counts, ensuring sufficient operational redundancy. Should predicted loads approach total available capacity, all available generators are brought online pre-emptively, safeguarding against blackout risks or power shortfalls.

Smooth transitions between generator combinations maintain operational stability:

- **Incremental Transition Enforcement:** If the recommended generator count increases, the system prevents immediate reduction back to a lower count by enforcing a configurable cooldown period; it's a mechanism that helps to stabilize the generator transitions and prevents frequent toggling to ensure long-term system stability.

Moreover, to support operator oversight and manual intervention, generator commands, status indicators, and key performance metrics are made accessible through a Human Machine Interface (HMI). This integration gives drillers real-time visibility into generator conditions, predicted capacity, load forecasts, and system decisions, enabling informed action in high-stakes scenarios.

Together, the safeguards, cost functions, and optimization logic described above form a robust, adaptive generator management system. Unlike traditional methods, this approach continuously balances competing objectives such as fuel efficiency, equipment longevity, and reliability, all while adapting to the dynamic demands of drilling. By proactively addressing issues like sudden derating, thermal stress, and load fluctuations, the system improves operational stability, lowers maintenance costs, and ensures uninterrupted power delivery in even the most challenging conditions.

Results & Analysis

A core insight of this work is the **critical importance—and versatility—of real-time generator capacity estimation** in modern drilling operations. Historically, most generator management systems have relied on rated capacities or assumed static derate margins, often set conservatively to account for possible degradation or environmental factors. However, as shown in [Figures 3](#) and [12](#), **there is a wide disparity**

between the rated capacity and the actual, real-time capacity of generators—not only between different rigs with identically rated engines, but also for the same generator over time due to aging, environment, or operational stress.

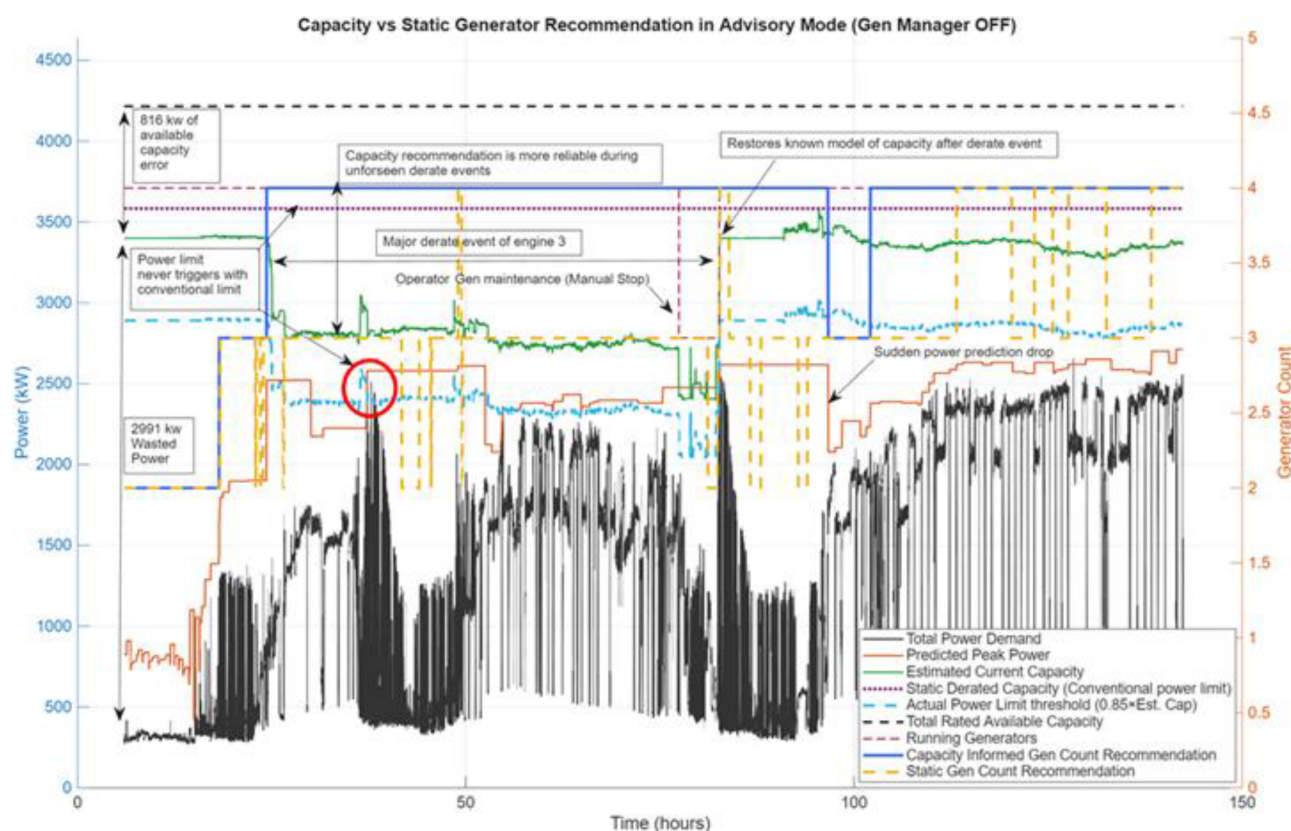


Figure 12—Comparison of Capacity-Informed and Static Generator Recommendations during an advisory-mode period (gen. manager off). The capacity recommendation (blue, solid) dynamically adapts to changes in available engine capacity, including sudden derate and maintenance events, while the conventional static approach (yellow, dashed) overestimates available power, risking blackouts. Additionally, the real power-limiting threshold (cyan) shows cases where conventional power limiting would never trigger, and should have.

This disparity is driven by a host of real-world factors: air filter restrictions, "cool mode" behavior, sensor calibration drift, accumulation of engine wear, fuel limitations, equipment differences (eg. Radiator Fans), and certain operational factors. Across multiple rigs, we observed derates exceeding 50% from nameplate ratings (Figures 5,12,13), with significant intra-unit variation even over the course of a single well cycle. Relying on static derate tables or nominal capacities—no matter how conservative—simply cannot capture this dynamic behavior, leaving operations vulnerable to two costly outcomes: **unplanned blackouts** (when capacity is overestimated), and **chronic inefficiency** (when capacity is underestimated and excess generation is kept online).

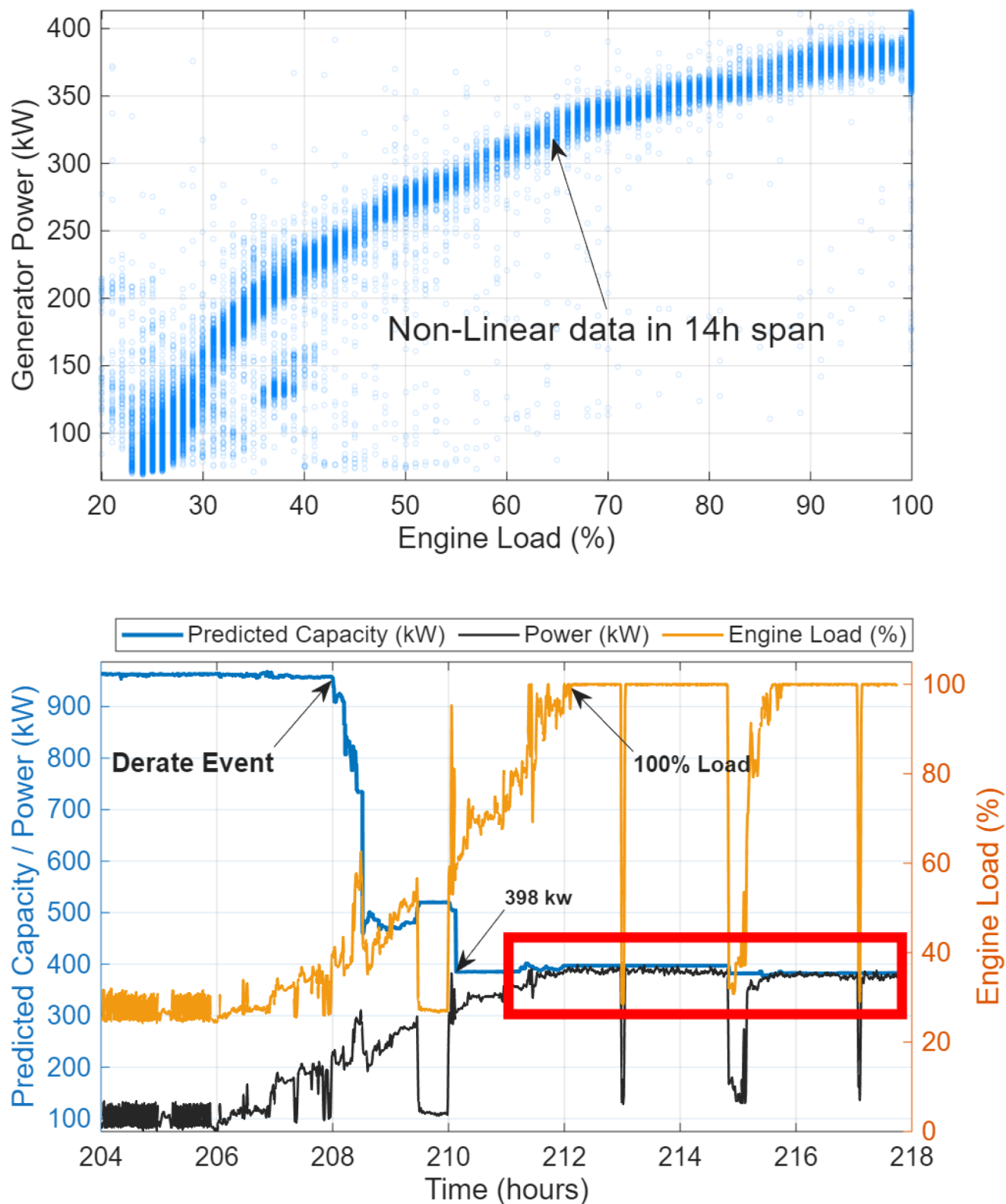


Figure 13—Validation of the IWLSR capacity estimation algorithm during a 14-hour operational window. (Top) Scatter plot of Generator power versus engine load highlights a period of non-linear behavior. (Bottom) Predicted capacity (blue), measured power (black), and engine load (orange) shows the estimator tracking through a sudden derate event and accurately converging to within ± 10 kW of true power at 100% load. The red box highlights the steady-state interval after derate, where estimated capacity matches actual generator output, confirming the robustness and accuracy of the approach under real-world conditions.

Beyond its central role in generator management, real-time capacity estimation also enables multiple critical applications across the rig's power system:

- **Power Limiting and Blackout Prevention:** Most drilling rigs implement power-limiting schemes to pre-empt blackouts—typically by phasing back equipment loads (e.g., top drive, mud pumps) when measured demand approaches assumed capacity. However, these limits are almost always set using rated or statically derated power values. As a result, in severe derate conditions, **safety interlocks may never trigger before a blackout occurs**. Integrating live capacity estimation into power limiting logic allow earlier, more accurate interventions, improving both operational safety and crew protection.
- **Hybridization and Energy Storage:** For rigs equipped with energy storage systems (ESS), **accurate generator capacity estimates are essential for effective charge/discharge scheduling** and reliable state-of-charge assessment. Real-time knowledge of true available power ensures that the ESS can buffer peaks and valleys with greater confidence, maximizing both fuel efficiency and system resilience.
- **Fleet Management and Efficiency:** When multiple generators are synchronized, severe derating of even a single engine can render the entire group less efficient or even cause total available output to decrease with more engines online. Real-time capacity estimates allow the system to dynamically select the most effective generator set (sometimes running fewer, healthier units for more power), preventing "weakest-link" limitations and supporting optimal fuel use.
- **Proactive maintenance:** Field experience has proven that the capacity estimate is a prime indicator of generator health, allowing for proactive maintenance to be performed. On numerous occasions, potential black out events were prevented by simply noticing that the capacity was starting to degrade, usually linked to air filter restrictions, but in one case, insufficient fuel was revealed hours in advance. Moreover, it helped identify equipment issues such as an incorrect thermostat that was putting several engines in cool mode intermittently, as seen in [Figure 7](#).

The adaptive estimator delivers all these benefits by providing highly accurate, continuously updated capacity estimates. [Figure 12](#) demonstrates that, even during sudden derate events and nonlinear transients, the estimator tracks true capacity to within ± 10 kW of delivered power at 100% load. This robustness holds across regimes, confirming both the accuracy and practical reliability of the approach. We found three major advantages in using it:

- **Improved robustness:** The risk of unexpected overloads and blackouts is minimized because available power is continuously and accurately tracked, even as individual units derate due to mechanical, thermal, or environmental stressors.
- **Increased fuel efficiency and reduced emissions:** As shown in [Figures 13a](#) and [14](#), generator efficiency and emissions profiles are highly nonlinear—operating generators at higher loads is dramatically more fuel-efficient and produces lower NO_x and greenhouse gases per unit of energy delivered. For instance, running two engines at 50% load is substantially "cleaner" than running four at 25%. The new management algorithm, by reliably quantifying available capacity, can safely reduce the number of online units, maximizing efficiency without risking undercapacity events.
- **Reduced wear and improved equipment life:** [Figures 14](#) and [5](#) show that traditional management approaches consistently result in significant unused power, as generators are kept online for redundancy or out of fear of derating events. With accurate generator management, aggregate run hours can be reduced by up to 36% ([Figure 14](#)), directly reducing maintenance intervals and extending equipment life.

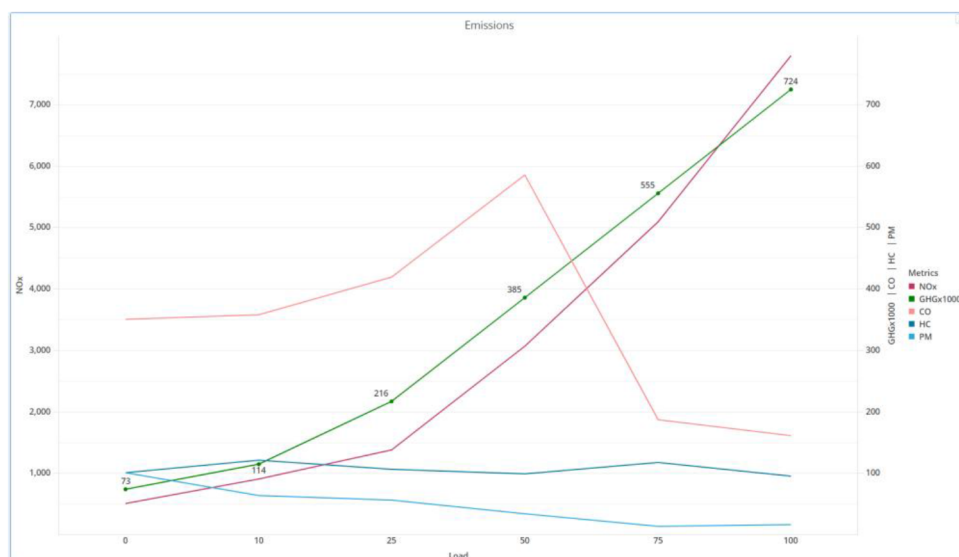


Figure 13a—Engine emissions versus generator load, illustrating the nonlinear relationship between load and pollutant output. Both greenhouse gas (GHG) emissions and NO_x increase with load, but not in direct proportion—per-unit emissions are significantly lower at higher loads, especially above 50%. This means that running fewer generators at higher load (e.g., two at 50% each) is far more fuel- and emissions-efficient than running more generators at low load (e.g., four at 25%), directly supporting the optimization logic of smart generator management systems.

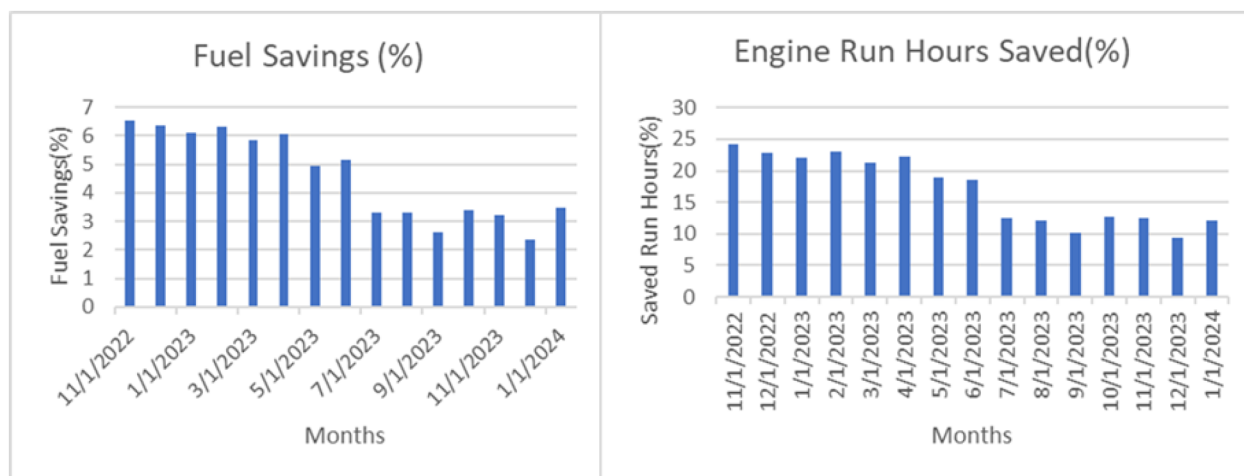


Figure 14—Monthly fuel savings (%) and engine run hours saved (%) from the deployment of smart generator management systems across 68 rigs and 725 wells over 14 months (adapted from [Koritala et al., 2023](#)). These results demonstrate consistent improvements, with average fuel savings ranging from 3–7% and reductions in engine run hours from 10–25%.

A field-wide study ([Koritala et al., 2023](#)) involving 68 rigs and 725 wells tracked performance after deployment of smart generator management systems. Over a 14-month period, monthly fuel usage savings ranged from 3% to 7%, with proportional reductions in CO₂ emissions. These improvements were consistent with expectations from our emissions/load analysis: more efficient generator loading, made possible by accurate live capacity estimation, directly reduces both fuel consumption and greenhouse gas output.

However, the primary shortcoming of prior smart management deployments was their reliance on fixed or statically derated capacities, as observed during field trials, which led to periods of either insufficient reserve or chronic overcapacity, undermining both reliability and economic benefits ([Fig 12](#)). The move to continuous, robust, and adaptive capacity estimation directly addresses this weakness.

While the deployment of real-time capacity estimation and adaptive generator management has delivered substantial gains in efficiency, equipment life, and reliability, our field data and operational experience reveal a fundamental upper bound to these improvements when using traditional engine-only architectures.

In particular, the **primary constraint to further reductions in fuel use and engine run hours is not average load, but rather the need to reliably meet high-magnitude peak power demands, such as those experienced during drilling "connection events."** Even with highly accurate capacity estimation, the system must keep sufficient generator capacity online to handle these transient peaks, limiting the extent to which overall engine hours can be reduced.

This finding suggests that, **to achieve step-change improvements beyond the 10% range observed here, integration of an energy storage system (ESS) or other forms of fast-responding reserve becomes essential.** An ESS can absorb short-term surges and smooth out load spikes, allowing the generator fleet to operate closer to average demand instead of sizing for rare worst-case peaks. Such hybrid architectures have the potential to deliver even greater reductions in fuel use, emissions, and wear, as well as improved blackout prevention.

However, the widespread adoption of ESS solutions in the drilling sector is currently constrained by cost, logistical complexity, and market maturity. Until these systems become economically viable for most rigs, **adaptive generator management using robust, real-time capacity estimation remains the most impactful and broadly applicable solution.** Nevertheless, our results suggest that as ESS technology matures, the combination of advanced generator management and storage will become the industry's most powerful tool for optimizing rig power systems.

Conclusions

Wide, dynamic **variations in real generator capacity are the norm** on modern drilling rigs, **making real-time capacity estimation essential for efficient reliable power management.** Static derate tables and fixed capacity assumptions often miss derates that can exceed 50%, due to air filter restrictions, cooling, fuel limitations, and wear.

Embedding live capacity estimation in generator management yields three core benefits:

Better blackout prevention: The system detects and responds to actual derates, avoiding overloads missed by static thresholds.

Greater efficiency and lower emissions: Field results show 3–7% monthly fuel and CO₂ savings, and up to 36% fewer gen. run hours.

Improved equipment life: Fewer unnecessary run hours and balanced loading reduce wear and enable earlier maintenance.

While adaptive management has delivered significant gains, **further optimization is constrained by the need to cover peak events.** Integrating energy storage systems (ESS) with real-time capacity estimation will be crucial to buffer surges and allow engines to run closer to average load.

Until ESS adoption is widespread, **robust capacity estimation is the most effective lever for operational reliability and cost savings** on today's rigs. As hybrid power solutions mature, combining ESS with adaptive generator management will define the next standard for drilling power systems.

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